An Advanced Lane-Keeping Assistance System with Switchable Assistance Modes

Yougang Bian, Jieyun Ding, Manjiang Hu, Qing Xu, Jianqiang Wang, and Keqiang Li

***Abstract*—Lane keeping is a key task in driving, and it plays an important role in staying safe while driving. However, conventional lane-keeping assistance systems (LKASs) have a limited level of automation, while fully autonomous lane-keeping systems have shortcomings regarding reliability. To address these issues, an advanced LKAS with two switchable assistance modes, namely, lane departure prevention mode and lane-keeping co-pilot mode, is proposed in this study. First, the system structure is constructed. Then, the functions and control strategies for the two assistance modes are defined. Next, the controller algorithms are designed using the learning-based model predictive control (LBMPC) method to compensate for modeling errors. Within the framework of LBMPC, an oracle is built to learn the unmodeled dynamics using an extended Kalman filter. Moreover, optimization problems are formulated to achieve the control objectives regarding driving safety and driver acceptance. Finally, driver-in-the-loop experiments carried out on a driving simulator prove that both assistance modes are effective. Furthermore, the lane-keeping co-pilot mode can help reduce driving burden in the sense of avoiding frequent steering correction.**

***Index Terms*—lane-keeping assistance system (LKAS), lane departure prevention (LDP), lane-keeping co-pilot, learning-based model predictive control (LBMPC).**

I. Introduction

As a typical type of advanced driving assistance systems, lane-keeping assistance systems (LKASs) aim to prevent unintended lane departures, so they are also called lane departure prevention (LDP) systems [1]. Various design methods have been proposed for LKASs using different types of actuators such as electric power steering (EPS) systems [2][3], steer-by-wire systems [4], differential braking systems [5], and combination solutions [6]. Recent advances in LKASs include learning-based design methods [7][8] and dataset-based evaluation and test methods [9]. In the field of civil engineering, the impact of road alignment on drivers' lane departure behaviors [10] has also been studied.

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Adaptability is a typical topic in the current research on LKASs, which can be reflected in two aspects. First, the decision of LKASs should adapt to drivers' intentions. Therefore, recognition of drivers' lane-change intentions is very important for LKASs. The commonly used methods for intention recognition include the hidden Markov model (HMM) [11], support vector machine (SVM) [12], and sparse Bayesian classification (SBC) [13]. Second, the control strategies in LKASs should be adjusted adaptively according to drivers' characteristics, which can be influenced by road types and traffic conditions. For example, Polychronopoulos *et al.* introduced a situation-adaptive LKAS in the PReVENT-SAFELANE project by considering situations in which the ego vehicle passes by trucks in adjacent lanes [14]. Batavia *et al.* designed a lane departure warning system (LDWS) [15] that takes into account drivers' lane-keeping characteristics and curve-cutting behaviors. An analysis of drivers' lane-keeping behaviors [16] further showed that drivers' lane-keeping characteristics are affected by driving conditions and can be effectively described using the concept of expected driving zone (EDZ).

Additionally, improving the level of automation in LKASs has become a promising research area due to the concept of autonomous vehicles. The conventional LKASs have fixed assistance modes; therefore, they can only assist drivers at a low level of automation over a short period of time. To improve the level of automation, Kim *et al.* proposed an active LKAS by combining a conventional LKAS and a fully automated lane-keeping system (LKS) [17]. However, since the driver is out of the control loop when using a fully automated LKS, his/her visual attention toward the forward roadway is obviously reduced [18]. This can lead to hazardous driving situations if the lane detection capabilities of LKSs are not absolutely reliable. To address this issue, Son *et al.* proposed the concept of predictive virtual lanes for normal lane-keeping operation when the lane detection module is momentarily unavailable [19]. Saito *et al.* designed a dual control scheme to perform safety control and identification of driver's state simultaneously [20]. However, these trial methods cannot handle all types of failure scenarios. Therefore, researchers have turned to transitional schemes to improve the level of automation in a progressive manner. One typical method used is the shared control assistance mode, or the "Co-Pilot" system, which has attracted much attention recently. For example, Anderson *et al.* studied the authority assignment strategy by assuming that the assistance controller was better than human

drivers when handling the most complicated driving tasks [21]. Thus, the weighting factor of the assistance controller was increased when the danger level increased. Boer *et al*. designed an index called the level of haptic authority to compute the weighted sum of the assistance control input and the driver input [22]. Drivers can manually adjust this level to confirm their choice of authority over the assistance controller. Sentouh *et al.* proposed a cooperative steering control system to avoid conflict between the intelligent controller and the driver in complex situations by predicting drivers' behaviors for control authority shifting [23].

Moreover, the design of LKAS controllers also needs further investigation to address the uncertainties and nonlinearities in vehicle dynamics, road conditions, and driver behaviors. For example, Saleh *et al*. applied the H2-Preview optimization in shared steering control to ensure the robustness regarding driver model uncertainties [24]. Gerdes *et al.* developed a Lyapunov-based bound within the energy theoretic framework to handle time-varying disturbances [25]. Among all these control methods, the model predictive control (MPC) is the best method suitable for handling constraints and nonlinearities. Therefore, it has been applied to vehicle control in numerous conducted studies. Liu *et al.* developed an MPC-based LKAS controller that can compute the robust invariant set to ensure bounded system state [26]. They further designed an unscented-transformation-based stochastic MPC controller for vehicle dynamics subject to nonlinearities to avoid a heavy workload [27]. However, MPC-based methods have heavy computational burden and may cause control conservativeness arising from model uncertainties. Therefore, the learning-based model predictive control (LBMPC) was proposed [28]. By introducing an oracle module that is updated based on an online learning approach and by incorporating the constraints developed by robust MPC [29], the LBMPC method can decouple robustness and optimal control performance, so that the performance can be enhanced without increasing computational complexity or sacrificing controller robustness. Therefore, we observe that the LBMPC can serve as a feasible method for the development of LKAS controllers.

To solve the problems mentioned above on the level of automation and control design, this study presents an advanced LKAS with two switchable assistance modes: a conventional LDP mode and a lane-keeping co-pilot (LK Co-Pilot) mode. The main contributions of this study include:

(1) Compared with traditional LKASs with fixed assistance modes, the proposed LKAS incorporates two switchable assistance modes. The LDP mode serves as the basic mode, while the LK Co-Pilot mode assists drivers to keep the vehicle within the EDZ without on/off state switching under normal conditions. Therefore, it is a feasible approach to improve the level of automation, as well as to keep the driver in the control loop.

(2) In the two assistance modes, LBMPC-based control algorithms are designed to address the nonlinearities and uncertainties in lateral control. These algorithms help improve the robustness of the proposed LKAS.

The rest of this paper is organized as follows. Section II introduces the system structure of the proposed LKAS. Section III defines the functions and control strategies for both assistance modes. Section IV describes the design of the LKAS controller algorithms within the framework of LBMPC. Driver-in-the-loop experiments on a driving simulator are conducted in Section V, while conclusions are presented in Section VI.

II. System Structure

The system structure of the proposed LKAS is shown in Fig. 1. The system consists of five modules:

(1) The environment sensing and vehicle state detection module obtains environment data, such as the positions of lane markings, road curvature, and road position, using onboard sensors and digital maps, and the state of the ego vehicle, such as velocity, steering angle, and yaw rate, through a vehicle CAN bus.

(2) The human-machine interface (HMI) module helps drivers select their preferred initial assistance mode: the LDP mode or the LK Co-Pilot mode. Each of the two modes has a separate strategy and a separate controller algorithm, whereas they share the same sensors and actuators.

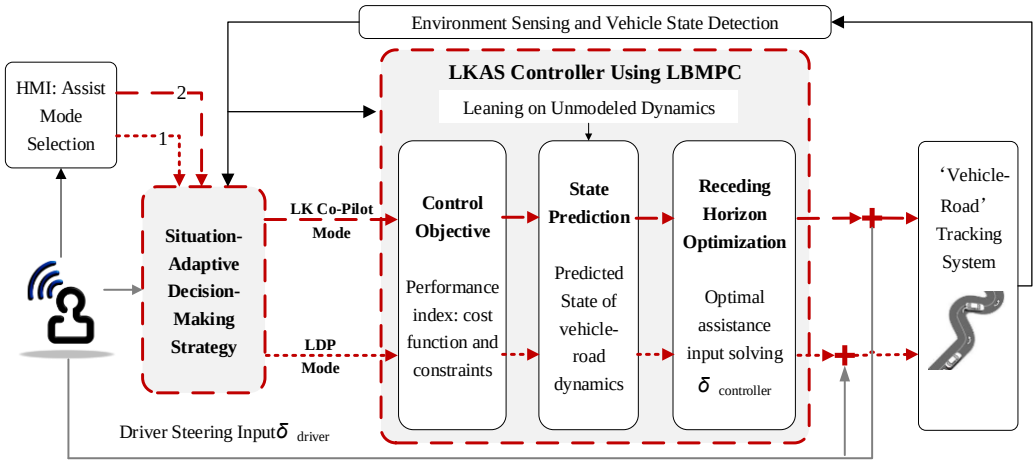


Fig. 1. System structure of the proposed LKAS

(3) The decision-making strategy module decides whether to provide assistance control depending on the selected mode. To ensure that the system is situation-adaptive, the strategy module integrates a lane-change intention-recognition algorithm [30] and a dynamically adjusted virtual lane border computation algorithm [16]. The first algorithm is important because it decides whether to provide assistance or not, while the second one determines threshold parameters using the EDZ.

(4) The LKAS controller takes charge of the vehicle depending on the decision made by the decision-making strategy module.

(5) The "vehicle-road" tracking system describes vehicle dynamics and the relationship between the vehicle and the road.

This study is focused on the design of the decision-making strategy module and the LKAS controller.

III. Mode Definition and Strategy Design

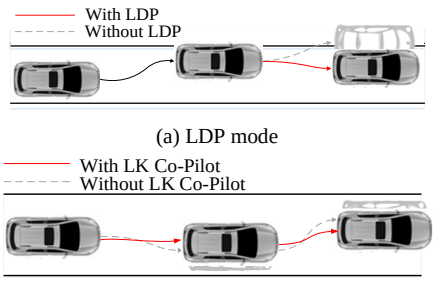
To ensure drivers' acceptance of the assistance control, dynamically adjusted virtual lane borders are used to calculate decision-making thresholds in both assistance modes. Moreover, the interaction between the driver and the LKAS is used to decide whether the assistance control will be executed or not.

A. LDP Mode

The LDP mode is the basic mode of the proposed LKAS. When lane departure is predicted or detected, the LDP function is activated to assist the steering control if no lane-change intention is detected, as depicted in Fig. 2.

In each control cycle, the on/off state of the LDP system must be set according to the interaction between the driver and the LKAS to avoid control conflict or excessive input. Moreover, to achieve the ideal control effect within few seconds, the thresholds must be adjusted by referring to the situation-adaptive virtual lane borders and the control objective must be adjusted based on the vehicle position. Instead of using the time-to-line-crossing to predict lane departure risks, as it is generally done in LDWS [31], here we use the lateral offset as the lane departure metric to determine whether or not to trigger the LDP mode, as it is done in [9].

Denote by e_y vehicle's lateral offset with respect to the lane center, and by Th_1, Th_2, and Th_3 three thresholds ($0 < Th_3 < Th_2 < Th_1$) to be used in decision making. As shown in Fig. 3, if $|e_y| > Th_1$ and no lane-change intention or proactive actions to steer back to the lane center are recognized or detected, then the LDP mode is activated. Thus, a two-stage strategy is proposed to adjust the main control objective (characterized by cost function J and weight matrix $\boldsymbol{Q}$ in optimization problem (15)): (1) in the first stage when $Th_2 < |e_y| \leq Th_1$, the main objective is to diminish the lateral offset, so we increase the weight of lateral offset (the 4th diagonal entry of $\boldsymbol{Q}$) in J (see TABLE I); (2) in the second stage when $Th_3 < |e_y| \leq Th_2$, in order to ensure a comfortable heading angle before shifting the control authority back to the driver, the main objective is to diminish the heading error, so we increase the weight of relative yaw angle (the 2nd diagonal entry of $\boldsymbol{Q}$) in J (see TABLE I).



(a) LDP mode

(b) LK Co-Pilot mode

Fig. 2. Functions of the LDP and LK Co-Pilot modes

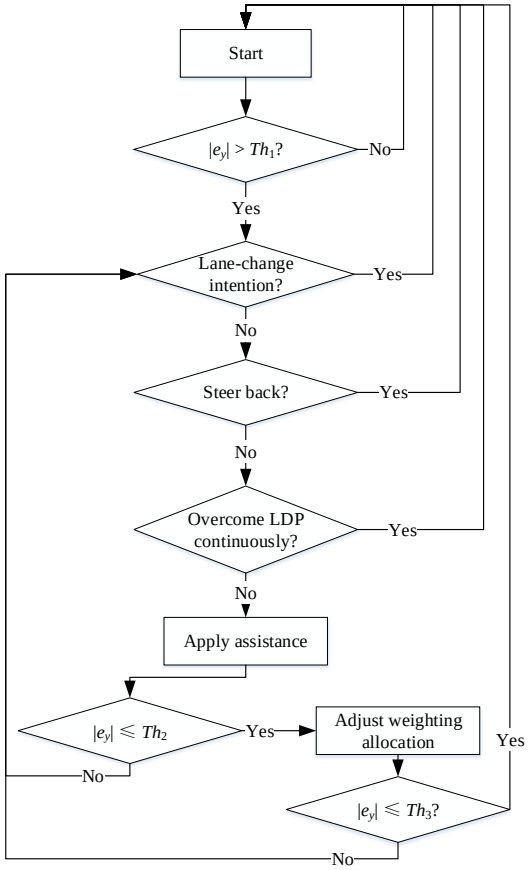


Fig. 3. Decision-making strategy of the LDP mode

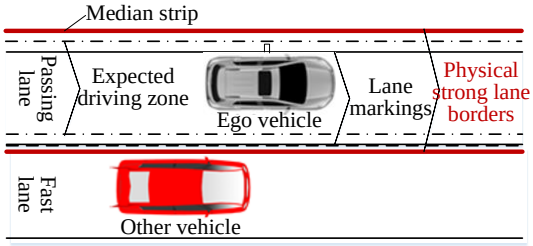


Fig. 4. Virtual lane border, physical lane border, and expected driving zone (EDZ)

Finally, the LDP mode is deactivated when $|e_y| \leq Th_3$. Moreover, let T_o denote a positive threshold to be designed; if the driver overcomes the assistance continuously for a time longer than T_o, then the LDP mode will also be deactivated. To further mitigate or avoid the discomfort arising from control authority shifting, we observe that the shared control method [23] may be an effective approach.

The threshold Th_1 corresponds to virtual lane borders and is related to two other thresholds (Th_{strong} and Th_{E}). The threshold Th_{strong} depends on physical strong lane borders and is designed to guarantee lateral safety. It is a little farther away from the lane center when compared with physical soft lane borders (Th_{soft}) that correspond to real lane borders marked with lane lines. The threshold Th_{E} describes the driver's accustomed driving area according to the EDZ. These two thresholds are shown in Fig. 4, where the ego vehicle drives in the left-most passing lane and another vehicle drives in the right adjacent lane. In this situation, the left physical strong lane border is the median strip, and the right physical strong lane border is about 0.2 m to the outside of the lane marking. Then the thresholds Th_1, Th_2, and Th_3 are given by

$$Th_1 = \min\{Th_{\text{strong}}, Th_{\text{E}}\}, \quad (1)$$

$$Th_2 = Th_1 - Th_{12}, \quad (2)$$

$$Th_3 = Th_1 - Th_{13}, \quad (3)$$

where Th_{12} and Th_{13} are positive thresholds to be designed.

In this study, we set $T_o = 1$ s, $Th_{12} = 0.15$ m, and $Th_{13} = 1$ m, which exhibit a satisfactory effect in simulator experiments. Optimization of these parameters needs a further study when developing a product LKAS.

B. LK Co-Pilot Mode

The LK Co-Pilot mode is the advanced mode. If the driver has no lane-change or switch-off intention, the assistance system helps the driver follow the expected trajectory by predicting the driver's dynamic steering input, as shown in Fig. 2. Therefore, it serves as a secondary pilot only.

As Fig. 5 shows, when the LK Co-Pilot mode is selected, it is always in the active state unless the lane-change intention is recognized, or lane markings are missed. Denote by $e_{y\text{EDZ}}$ the offset of the center of the EDZ with respect to the lane center, and by D_{offset} a positive threshold to be designed. Then, a two-stage strategy is proposed to adjust the main control objective (characterized by cost function J and weight matrix $\boldsymbol{Q}$ in optimization problem (21)): (1) in the first stage when $|e_y - e_{y\text{EDZ}}| > D_{\text{offset}}$, the weight of lateral offset (the 4th entry of $\boldsymbol{Q}$) is increased (see TABLE II); (2) in the second stage when $|e_y - e_{y\text{EDZ}}| \le D_{\text{offset}}$, the weight of lateral offset is decreased (see TABLE II) to enhance the driver's control feedback. In this study, we set $D_{\text{offset}} = 0.5$ m, which exhibits a satisfactory effect in simulator experiments. A rigorous threshold selection method needs a further study in the development of a product LKAS.

Note that if the lane-change intention is not recognized correctly in time, conflict may arise between the assistance system and the driver. In this case, according to the intention-recognition algorithm [30], the driver's continuous overcoming behavior will increase the possibility of correct intention recognition, so the assistance will be canceled to avoid conflict. Another feasible approach is the same as that employed in the LDP mode: if the driver continuously overcomes the assistance for a time longer than a preset threshold, the assistance will be switched off to avoid conflict. Therefore, the intention-recognition algorithm and the preset threshold deserve a further study to avoid or mitigate conflict when developing a product LKAS.

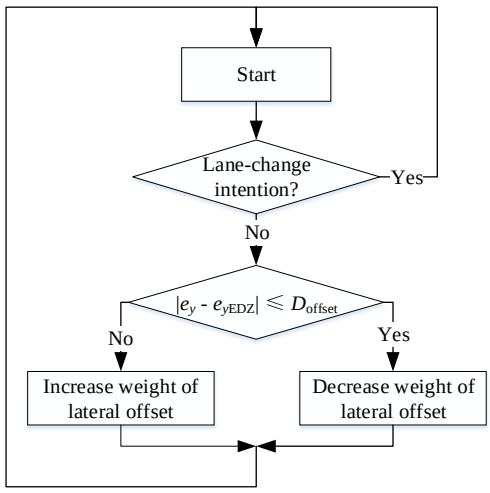


Fig. 5. Decision-making strategy of the LK Co-Pilot mode

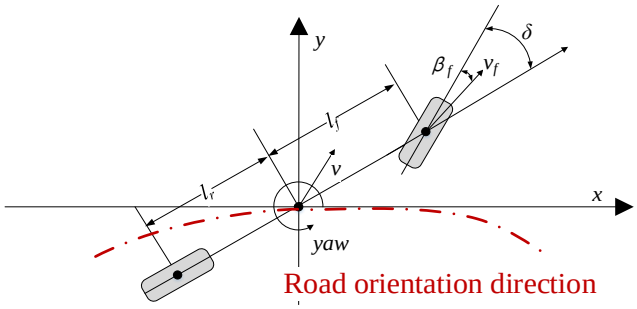


Fig. 6. Vehicle-road tracking system

IV. LKAS Controller Algorithm

The proposed LKAS controller algorithm is designed within the framework of LBMPC. First, a nominal model is built for the "driver-vehicle-road" system. Next, a linear oracle is built for learning the simplified unmodeled dynamics. Then, optimization problems are designed to formulate the control objectives and system constraints regarding safety, stability, and robustness. These optimization problems are solved in a receding-horizon manner to generate the desired control inputs.

A. Nominal System Model

The nominal model is an approximate linear time-invariant (LTI) model with bounded modeling error. The linear dynamics and the bounds of the unmodeled uncertainties are used to compute the constraint set, which certifies that the system states will converge robustly at the tube centered at the state generated by the LTI model.

The LTI model of the closed-loop "driver-vehicle-road" tracking system describes the basic linear dynamics of the system. It consists of a vehicle model, a "vehicle-road" model, and a driver model. As shown in Fig. 6, we use the following linear two-degree-of-freedom bicycle model as one part of the vehicle-road dynamic system:

$$\dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}u + \boldsymbol{E}\rho, \quad (4)$$

$$\boldsymbol{x}=\begin{bmatrix}\dot{\psi}\\ e_\psi\\ v_{yr}\\ e_y\end{bmatrix},\boldsymbol{A}=\begin{bmatrix}a_{11}&a_{12}&a_{13}&0\\ 1&0&0&0\\ a_{31}&a_{32}&a_{33}&0\\ 0&0&1&0\end{bmatrix},$$
$$\boldsymbol{B}=\begin{bmatrix}b_1\\ 0\\ b_3\\ 0\end{bmatrix},u=\delta,\boldsymbol{E}=\begin{bmatrix}0\\ -v\\ -v^2\\ 0\end{bmatrix},$$

where $\boldsymbol{x}$ is the system state, $\dot{\psi}$ is the yaw rate, e_ψ is the relative yaw angle, v_{yr} is the relative lateral speed, e_y is the lateral offset, u as the system input is the steering angle of the front wheel denoted by δ, v is the longitudinal speed, and ρ is the road curvature and is treated as an external input. The parameters in matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ are given by

$$\begin{gathered}a_{11}=-\frac{2C_fl_f^2+2C_rl_r^2}{I_zv},a_{12}=\frac{2C_fl_f-2C_rl_r}{I_z},\\ a_{13}=-\frac{2C_fl_f-2C_rl_r}{I_zv},a_{31}=-\frac{2C_fl_f-2C_rl_r}{mv},\\ a_{32}=\frac{2C_f+2C_r}{m},a_{33}=-\frac{2C_f+2C_r}{mv},\\ b_1=\frac{2C_fl_f}{I_z},b_3=\frac{2C_f}{m},\end{gathered}\tag{5}$$

where $l_f(l_r)$ is the distance between the front (rear) axle and the mass center, $C_f(C_r)$ is the cornering stiffness of the front (rear) tire, m and I_z are the mass and moment of inertia, respectively.

It is assumed that the driver adjusts the steering angle regarding the lateral offset error and the yaw angle error at the current position and the preview point, as illustrated in Fig. 7. Then, the simplified driver model can be given by

$$\begin{aligned}\delta_d&=k_ye_y+k_{\psi,1}e_\psi+k_{\psi,2}\Delta\psi_{\text{road}}\\ &=[0\quad k_{\psi,1}\quad 0\quad k_y]\boldsymbol{x}+k_{\psi,2}\Delta\psi_{\text{road}},\end{aligned}\tag{6}$$

where δ_d is the driver's input angle on the steering wheel, $\Delta\psi_{\text{road}}$ is the change in the road orientation angle at the preview point, k_y, $k_{\psi,1}$, and $k_{\psi,2}$ are the gain coefficients of the current lateral offset error, yaw angle error, and change in the road orientation angle, respectively.

By substituting (6) into (4), we obtain the following closed-loop system:

$$\begin{aligned}\dot{\boldsymbol{x}}&=\left(\boldsymbol{A}+\boldsymbol{B}G_{\text{fw/sw}}\boldsymbol{G}_d\right)\boldsymbol{x}+\boldsymbol{B}G_{\text{fw/sw}}u_c+\boldsymbol{E}\rho+\boldsymbol{E}_d,\\ \boldsymbol{G}_d&=[0\quad k_{\psi,1}\quad 0\quad k_y],\\ \boldsymbol{E}_d&=\boldsymbol{B}G_{\text{fw/sw}}k_{\psi,2}\Delta\psi_{\text{road}},\end{aligned}\tag{7}$$

where $G_{\text{fw/sw}}$ is the transfer ratio from the steering wheel angle to the front wheel angle, u_c is the assistant steering wheel angle. Based on system (7), a discrete model with bounded disturbances is further obtained for predictive usage. Note that larger disturbances will induce higher levels of conservativeness, which will result in a narrower range of feasible solutions. Therefore, to properly diminish the range of disturbances, we consider the following three LTI models that employ different nominal vehicle longitudinal speeds:

$$\begin{aligned}&\boldsymbol{x}_{k+1}=\boldsymbol{A}_{\text{DVR},i}\boldsymbol{x}_k+\boldsymbol{B}_{\text{DVR},i}u_k+\boldsymbol{E}_{\text{DVR}}+\boldsymbol{d}_k,\\ &\text{if}\quad v<80,\quad\text{then } i=1,\bar{v}=70,\\ &\text{else if } v<100,\quad\text{then } i=2,\bar{v}=90,\\ &\text{else if } v<120,\quad\text{then } i=3,\bar{v}=110,\end{aligned}\tag{8}$$

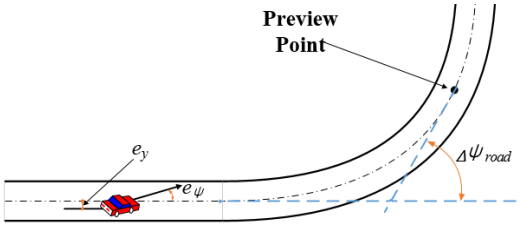


Fig. 7. Three factors in the simplified driver model

where $\boldsymbol{A}_{\text{DVR},i}$, $\boldsymbol{B}_{\text{DVR},i}$, and $\boldsymbol{E}_{\text{DVR}}$ are the discretized parameter matrices of system (7), $\bar{v}$ is the nominal speed, $\boldsymbol{d}_k$ is the state error arising from unmodeled dynamics. Here, the unmodeled dynamics include the unmeasurable vehicle parameters, indeterminate tire characteristics, stochastic driver behaviors, and environment recognition errors. Therefore, $\boldsymbol{d}_k$ belongs to the Minkowski sum ($\oplus$) of the bounds of these unmodeled dynamics:

$$\boldsymbol{D}_{\text{DVR}}\coloneqq\boldsymbol{D}_{\text{DVR_}d}\oplus\boldsymbol{D}_{\text{DVR_}v}\oplus\boldsymbol{D}_{\text{DVR_}r},\tag{9}$$

where $\boldsymbol{D}_{\text{DVR_}d}$, $\boldsymbol{D}_{\text{DVR_}v}$, and $\boldsymbol{D}_{\text{DVR_}r}$ are the bounds of the disturbances generated by the driver modeling error, vehicle modeling error, and road recognition error, respectively. See [32] for more details on system (8).

B. Oracle Design and Learning

A parametric oracle is built to learn the unmodeled dynamics online. The oracle and the nominal model together constitute a complete model, called *the oracle model*, to better describe the real system. If the oracle's algebraic values lie within the bounds of disturbances calculated beforehand and the states predicted by the nominal model satisfy the constraints of the robust invariant set, then the optimal solution that minimizes the cost function of the states predicted by the oracle model will be feasible. Meanwhile, the optimal solution can also lead to better control performance.

The unmodeled dynamics is generally nonlinear and difficult to be formulated using a physical mechanism for reasons such as stochastic driver behaviors. However, within a short period of time, *e.g.*, one control cycle, the unmodeled dynamics can be regarded as approximately linear. Therefore, we can use the following linear oracle to describe the unmodeled dynamics:

$$\boldsymbol{O}_k(\boldsymbol{x},u)=\boldsymbol{F}\boldsymbol{x}+\boldsymbol{H}u+\boldsymbol{G},\tag{10}$$
$$\boldsymbol{F}=\begin{bmatrix}\beta_1&\beta_2&\beta_3&\beta_4\\ \beta_5&\beta_6&\beta_7&\beta_8\\ \beta_9&\beta_{10}&\beta_{11}&\beta_{12}\\ \beta_{13}&\beta_{14}&\beta_{15}&\beta_{16}\end{bmatrix},\boldsymbol{H}=\begin{bmatrix}\beta_{17}\\ \beta_{18}\\ \beta_{19}\\ \beta_{20}\end{bmatrix},\boldsymbol{G}=\begin{bmatrix}\beta_{21}\\ \beta_{22}\\ \beta_{23}\\ \beta_{24}\end{bmatrix}.$$

In oracle (10), each of the matrices consists of floating parameters. These floating parameters will be estimated using an extended Kalman filter (EKF) because this method can inhibit noises and it has a low computational load and a low storage capacity demand [15].

By augmenting the state of system (8) with the floating parameters of matrices $\boldsymbol{F}$, $\boldsymbol{H}$, and $\boldsymbol{G}$ in oracle (10), we get the following new system:

$$\boldsymbol{x}_{\text{aug},k+1}=\begin{bmatrix}\boldsymbol{x}_{k+1}\\ \beta_{1,k+1}\\ \vdots\\ \beta_{24,k+1}\end{bmatrix}=f\left(\boldsymbol{x}_{\text{aug},k}\right)\tag{11}$$

$$= \begin{bmatrix} (\boldsymbol{A}_{\mathrm{DVR},k} + \boldsymbol{F}_k)\boldsymbol{x}_k + (\boldsymbol{B}_{\mathrm{DVR},k} + \boldsymbol{H}_k)u_k + \boldsymbol{E}_{\mathrm{DVR}} + \boldsymbol{G}_k \\ \beta_{1,k} \\ \vdots \\ \beta_{24,k} \end{bmatrix},$$

$$\boldsymbol{y}_{\mathrm{aug},k+1} = h(\boldsymbol{x}_{\mathrm{aug},k+1}) = \begin{bmatrix} 1 & 0 & 0 & 0 & \boldsymbol{0}_{1\times 24} \\ 0 & 0 & 0 & 1 & \boldsymbol{0}_{1\times 24} \end{bmatrix} \boldsymbol{x}_{\mathrm{aug},k+1},$$

where $\boldsymbol{x}_{\mathrm{aug}}$ and $\boldsymbol{y}_{\mathrm{aug}}$ are the system state and output, respectively. Moreover, the output $\boldsymbol{y}_{\mathrm{aug}}$ consists of the absolute yaw rate and lateral offset, which can be measured directly.

Note that system (11) is nonlinear with respect to the augmented system state $\boldsymbol{x}_{\mathrm{aug}}$. By taking the first-order partial derivatives of $f(\cdot)$ and $h(\cdot)$ in system (11) with respect to $\boldsymbol{x}_{\mathrm{aug}}$, we have:

$$\begin{aligned} \Phi_{k+1} &= \left.\frac{\partial f(\boldsymbol{x},k)}{\partial \boldsymbol{x}}\right|_{\boldsymbol{x}=\boldsymbol{x}^+_{\mathrm{aug},k}}, \\ \mathrm{H}_{k+1} &= \left.\frac{\partial h(\boldsymbol{x},k+1)}{\partial \boldsymbol{x}}\right|_{\boldsymbol{x}=\boldsymbol{x}^-_{\mathrm{aug},k+1}}. \end{aligned} \tag{12}$$

Thus, $\boldsymbol{x}_{\mathrm{aug}}$ can be estimated using the EKF algorithm, which consists of two steps. The first step predicts the system state in the next sampling interval preliminarily and propagates the cross-covariance matrix $\boldsymbol{P}_k$:

$$\begin{cases} \boldsymbol{x}^-_{k+1} = f(\boldsymbol{x}^+_k, k) + \Phi_{k+1} \cdot d\boldsymbol{x}_{\mathrm{aug},k}, \\ \boldsymbol{P}^-_{k+1} = \Phi_k \boldsymbol{P}^+_k \Phi^T_k + \boldsymbol{Q}_{k+1}, \\ \boldsymbol{y}^-_{k+1} = h(\boldsymbol{x}^-_{k+1}, k+1) + \mathrm{H}_{k+1} \cdot d\boldsymbol{x}_{\mathrm{aug},k+1}, \end{cases} \tag{13}$$

where $d\boldsymbol{x}_{\mathrm{aug},k}$ is the disturbance located within the linearization neighbor region, $\boldsymbol{Q}_{k+1}$ is the covariance of the parameter noise arising from system uncertainties, and $\boldsymbol{y}^-_{k+1}$ is the preliminarily estimated output. The second step conducts feedback correction with the error between $\boldsymbol{y}_{k+1}$ and $\boldsymbol{y}^-_{k+1}$. The Kalman gain $\boldsymbol{K}_{k+1}$, corrected state $\boldsymbol{x}^+_{k+1}$, and covariance $\boldsymbol{P}^+_{k+1}$ are updated as follows:

$$\begin{cases} \boldsymbol{K}_{k+1} = \boldsymbol{P}^-_{k+1}\mathrm{H}^T_{k+1}(\mathrm{H}_{k+1}\boldsymbol{P}^-_{k+1}\mathrm{H}^T_{k+1} + \boldsymbol{R}_{k+1})^{-1}, \\ \boldsymbol{x}^+_{k+1} = \boldsymbol{x}^-_{k+1} + \boldsymbol{K}_{k+1}(\boldsymbol{y}_{k+1} - \boldsymbol{y}^-_{k+1}), \\ \boldsymbol{P}^+_{k+1} = (\boldsymbol{I} - \boldsymbol{K}_{k+1}\mathrm{H}_{k+1})\boldsymbol{P}^-_{k+1}, \end{cases} \tag{14}$$

where $\boldsymbol{R}_{k+1}$ is the covariance of the measurement noise.

C. Optimization Problem for LDP Mode

Using the oracle model given above, the optimization problem for the LDP mode is formulated as follows:

$$\min_{\{c_0,c_1,\ldots,c_{N-1}\},\varepsilon} J = \sum_{i=0}^{N-1} \left(\|\tilde{\boldsymbol{x}}_i - \bar{\boldsymbol{x}}_{\mathrm{s}}\|^2_{\boldsymbol{Q}} + \|\breve{u}_i - \bar{u}_{\mathrm{s}}\|^2_r \right) + \|\tilde{\boldsymbol{x}}_N - \bar{\boldsymbol{x}}_{\mathrm{s}}\|^2_{\boldsymbol{P}} + w_\varepsilon \varepsilon^2$$

s.t.: $k = 0,1,\ldots,N-1,$

(a) $\tilde{\boldsymbol{x}}_0 = \boldsymbol{x}_0, \bar{\boldsymbol{x}}_0 = \boldsymbol{x}_0$

(b) $\tilde{\boldsymbol{x}}_{k+1} = (\boldsymbol{A}_{\mathrm{DVR}} + \boldsymbol{F})\tilde{\boldsymbol{x}}_k + (\boldsymbol{B}_{\mathrm{DVR}} + \boldsymbol{H})\breve{u}_k + \boldsymbol{E}_{\mathrm{DVR}} + \boldsymbol{G}$

(c) $\bar{\boldsymbol{x}}_{k+1} = \boldsymbol{A}_{\mathrm{DVR}}\bar{\boldsymbol{x}}_k + \boldsymbol{B}_{\mathrm{DVR}}\bar{u}_k + \boldsymbol{E}_{\mathrm{DVR}}$

(d) $\breve{u}_k = \bar{u}_k + c_k, \bar{u}_k = \boldsymbol{K}\bar{\boldsymbol{x}}_k$ (15)

(e) $\breve{u}_k \in \boldsymbol{\mathcal{U}}$

(f) $(\bar{\boldsymbol{x}}^T_k \quad \breve{u}_k)^T \in \boldsymbol{\mathcal{XU}}$

(g) $\Delta\breve{u}_k \in \Delta\boldsymbol{\mathcal{U}}$

(h) $\bar{\boldsymbol{x}}_N \in \boldsymbol{\mathcal{W}}$

where $\boldsymbol{x}$ and u are the system state and assistant control input, respectively; the symbols with subscript 's' represent the desired variables in the stable state; the symbols with numerical subscripts ranging from 0 to N represent variables in the current sampling interval and the prediction horizon with length N; the states predicted using oracle model (15)-(b) have superscripts of "~," while the states predicted using nominal model (15)-(c) have superscripts of "–"; "$\breve{u}$" is the input used in both models; and $\boldsymbol{Q}$, $\boldsymbol{P}$, and r are weight matrices or coefficients.

The cost function in (15) consists of three parts:

(1) The first part is the stage cost consisting of the weighted state and input errors. This cost drives the system closer to the stable state of the nominal system. The stable state is given by:

$$(\boldsymbol{I} - \boldsymbol{A}_{\mathrm{DVR}})\bar{\boldsymbol{x}}_{\mathrm{s}} = \boldsymbol{B}_{\mathrm{DVR}}\bar{u}_{\mathrm{s}} + \boldsymbol{E}_{\mathrm{DVR}}, \tag{16}$$

where $\bar{u}_{\mathrm{s}}$ is selected to ensure that the vehicle stays at the center of the EDZ in the stable state, which is the main control objective.

(2) The second part is the terminal cost consisting of the weighted terminal state error. Similar to the optimal control, the weights of the errors satisfy the following discrete algebraic Riccati equation (ARE):

$$\begin{aligned} (\boldsymbol{A}_{\mathrm{DVR}} + \boldsymbol{B}_{\mathrm{DVR}}\boldsymbol{K})^T\boldsymbol{P}(\boldsymbol{A}_{\mathrm{DVR}} + \boldsymbol{B}_{\mathrm{DVR}}\boldsymbol{K}) - \boldsymbol{P} \\ = -(\boldsymbol{Q} + r\boldsymbol{K}^T\boldsymbol{K}), \\ \boldsymbol{K} = -(r + \boldsymbol{B}^T_{\mathrm{DVR}}\boldsymbol{P}\boldsymbol{B}_{\mathrm{DVR}})^{-1}\boldsymbol{B}^T_{\mathrm{DVR}}\boldsymbol{P}\boldsymbol{A}_{\mathrm{DVR}}. \end{aligned} \tag{17}$$

(3) The third part is the penalty on the relaxation factor ε.

As (15)-(d) shows, the control input in the predictive horizon consists of two parts. The first part '$\bar{u}_k$' is a feedback control with feedback gain $\boldsymbol{K}$, while the second part c_k is an additional control part. The sum of the two parts should satisfy the input constraint and the input gradient constraint. The feedback gain is pre-determined by ARE (17) to guarantee asymptotic stability of the nominal system. Therefore, the second part should be optimized by minimizing the cost function.

As shown in (15)-(e)~(g), due to the dynamics constraints, drivers' acceptance, and lateral stability constraints, the control input u, slip angles of both the front and the rear tires α, and gradient of control input Δu should belong to specific constraint set represented by the following linear matrix inequalities:

$$\boldsymbol{\mathcal{U}}: \boldsymbol{F}_u u < \boldsymbol{h}_u, \boldsymbol{F}_u = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \boldsymbol{h}_u = \begin{bmatrix} u_{\max} \\ -u_{\min} \end{bmatrix}; \tag{18}$$

$$\Delta\boldsymbol{\mathcal{U}}: \boldsymbol{F}_{\Delta u}\Delta u < \boldsymbol{h}_{\Delta u}, \boldsymbol{F}_{\Delta u} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \boldsymbol{h}_{\Delta u} = \begin{bmatrix} \Delta u_{\max} \\ -\Delta u_{\min} \end{bmatrix}; \tag{19}$$

$$\boldsymbol{\mathcal{XU}}: \boldsymbol{F}_{xu}\begin{bmatrix} \boldsymbol{x} \\ u \end{bmatrix} < \boldsymbol{h}_{xu}, \tag{20}$$

$$\boldsymbol{F}_{xu} = \begin{bmatrix} -l_{\mathrm{f}}/v & 1 + G_{\mathrm{fw/sw}}k_{\psi,1} & -1/v & G_{\mathrm{fw/sw}}k_y & G_{\mathrm{fw/sw}} \\ l_{\mathrm{f}}/v & 1 + G_{\mathrm{fw/sw}}k_{\psi,1} & 1/v & G_{\mathrm{fw/sw}}k_y & G_{\mathrm{fw/sw}} \\ l_{\mathrm{r}}/v & 1 & 0 & 0 & 0 \\ -l_{\mathrm{r}}/v & -1 & 0 & 0 & 0 \end{bmatrix},$$

$$\boldsymbol{h}_{xu} = \begin{bmatrix} (1+\varepsilon)\alpha_{\mathrm{fmax}} - G_{\mathrm{fw/sw}}k_{\psi,2}\Delta\psi_{\mathrm{road}} \\ -(1+\varepsilon)\alpha_{\mathrm{fmin}} + G_{\mathrm{fw/sw}}k_{\psi,2}\Delta\psi_{\mathrm{road}} \\ (1+\varepsilon)\alpha_{\mathrm{rmax}} \\ -(1+\varepsilon)\alpha_{\mathrm{rmin}} \end{bmatrix};$$

where α_{fmax} and α_{fmin} (α_{rmax} and α_{rmin}) are the maximum and minimum slip angles of the front (rear) wheels, respectively.

As shown in (15)-(h), the system state is restricted to an approximate disturbance invariant set $\boldsymbol{\mathcal{W}}$ that can be computed

using the method given in [28].

The parameters in (15) are listed in TABLE I. The sampling time and the length of the predictive horizon are selected based on the computing capability of the simulator to balance the control performance and the real-time performance. The minimum and maximum slip angles are selected according to the statistics of natural driving data. The minimum (maximum) control input, gradient of control input, and the weightings are selected by manual parameter tuning to achieve satisfactory performance.

D. Optimization Problem for LK Co-Pilot Mode

Similarly, the optimization problem for the LK Co-Pilot mode is formulated as follows:

$$\min_{\{c_0,c_1,\dots,c_{N-1}\},\varepsilon} J=\sum_{i=0}^{N-1}\left(\|\tilde{\boldsymbol{x}}_i-\bar{\boldsymbol{x}}_s\|_{\boldsymbol{Q}}^2+\|\breve{u}_i-\breve{u}_s\|_r^2\right) +\|\tilde{\boldsymbol{x}}_N-\bar{\boldsymbol{x}}_s\|_{\boldsymbol{P}}^2+w_\varepsilon\varepsilon^2$$

s.t.: $k=0,1,\dots,N-1$,

(a) $\tilde{\boldsymbol{x}}_0=\boldsymbol{x}_0,\bar{\boldsymbol{x}}_0=\boldsymbol{x}_0$

(b) $\tilde{\boldsymbol{x}}_{k+1}=(\boldsymbol{A}_{\mathrm{DVR}}+\boldsymbol{F})\tilde{\boldsymbol{x}}_k+(\boldsymbol{B}_{\mathrm{DVR}}+\boldsymbol{H})\breve{u}_k+\boldsymbol{E}_{\mathrm{DVR}}+\boldsymbol{G}$

(c) $\bar{\boldsymbol{x}}_{k+1}=\boldsymbol{A}_{\mathrm{DVR}}\bar{\boldsymbol{x}}_k+\boldsymbol{B}_{\mathrm{DVR}}\breve{u}_k+\boldsymbol{E}_{\mathrm{DVR}}$

(d) $\breve{u}_k=\bar{u}_k+c_k,\bar{u}_k=\boldsymbol{K}\bar{\boldsymbol{x}}_k$ (21)

(e) $\bar{\boldsymbol{x}}_{k+1}\in\boldsymbol{\mathcal{X}},\bar{\boldsymbol{x}}_1\in\boldsymbol{\mathcal{X}}\ominus\boldsymbol{D}_{\mathrm{DVR}}$

(f) $\breve{u}_k\in\boldsymbol{\mathcal{U}}$

(g) $(\bar{\boldsymbol{x}}_k^T\quad\breve{u}_k)^T\in\boldsymbol{\mathcal{XU}}$

(h) $\Delta\breve{u}_k\in\Delta\boldsymbol{\mathcal{U}}$

(i) $\bar{\boldsymbol{x}}_N\in\boldsymbol{\mathcal{W}}$

where $\ominus$ denotes Minkowski difference.

Compared with optimization problem (15), optimization problem (21) has an additional constraint (21)-(e) to ensure that the vehicle's lateral position remains within a safe region.

The parameters in optimization problem (21) are listed in TABLE II. The minimum (maximum) control input, gradient of control input, vehicle lateral offset, and the weightings are selected by manual parameter tuning to achieve satisfactory performance.

TABLE I
PARAMETERS OF THE LDP MODE CONTROL LAW

Symbol	Meaning	Value
T	sampling time	100 ms
N	length of predictive horizon	10
$\alpha_{\min}/\alpha_{\max}$	minimum/maximum slip angle	−4 °/4 °
$u_{\min}/u_{\max}$	minimum/maximum control input	−5 °/5 °
$\Delta u_{\min}/\Delta u_{\max}$	minimum/maximum gradient of control input	−5 °/5 °
$\boldsymbol{Q}$	weighting matrix of system state (1st stage/2nd stage)	diag{0,1,0,10}/ diag{0,10,0,5}
r	weight of control input	1
w_ε	weight of relaxation factor	50

TABLE II
PARAMETERS OF THE LK CO-PILOT MODE CONTROL LAW

Symbol	Meaning	Value
T	sampling time	100 ms
N	length of predictive horizon	10
$\alpha_{\min}/\alpha_{\max}$	minimum/maximum slip angle	−4 °/4 °
$u_{\min}/u_{\max}$	minimum/maximum control input	−10°~−20°/10°~20°
$\Delta u_{\min}/\Delta u_{\max}$	minimum/maximum gradient of control input	−10 °/10 °
$e_{\mathrm{ymin}}/e_{\mathrm{ymax}}$	minimum/maximum vehicle lateral offset	−1 m/1 m
$\boldsymbol{Q}$	weighting matrix of system state (1st stage/2nd stage)	diag{20,10,0,20}/ diag{50,10,0,5}
r	weight of control input	1
w_ε	weight of relaxation factor	50

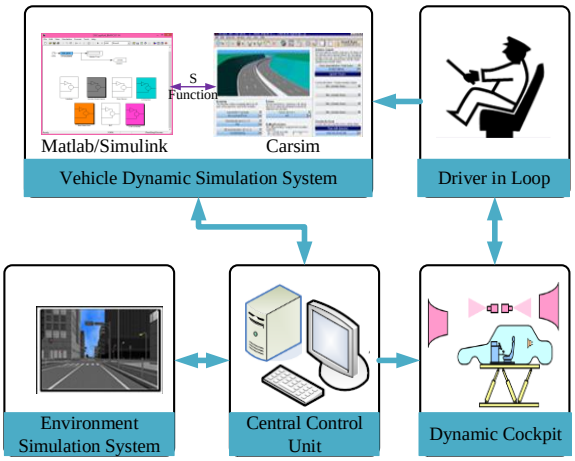


Fig. 8. Structure of the driving simulator

V. SIMULATOR EXPERIMENTS AND ANALYSIS

The performance of LKASs has strong relationship with human drivers' stochastic behaviors. Therefore, driver-in-the-loop experiments were conducted on a driving simulator to validate the effectiveness of the proposed LKAS. As Fig. 8 shows, the driving simulator contains four parts:

(1) The vehicle dynamic simulation system contains the vehicle model configured in Carsim and the proposed LKAS constructed in Matlab/Simulink.

(2) The environment simulation system defines the surrounding environment, including the road, roadside buildings, and other vehicles.

(3) The dynamic cockpit is equipped with a steering wheel, a throttle pedal, and a brake pedal. All these actuators have force feedback capabilities.

(4) The central control unit takes control of all other subsystems mentioned above.

The driving simulator is assumed to have reliable sensors and actuators. For more details on the driving simulator and the system model, the reader is referred to [32].

To implement the proposed LKAS on the driving simulator, we convert the designed optimization problems to quadratic programming (QP) problems so that they can be solved using the *fmincon* function in MATLAB. The time cost to solve these problems is less than the sampling time, so control inputs can be generated in real time. We note that other types of efficient computation methods for MPC [33][34] can also be used.

A 28-kilometer-long ring road consisting of both straight and curved parts was used in the experiment. A male driver who tends to drive toward the left side of the lane participated in the experiment. In addition, both lane-keeping and lane-change scenarios were designed for the experiment.

A. Controller Performance

First, we evaluated the performance of the proposed LBMPC-based controller in comparison with a MPC-based controller. These two controllers are both built based on

optimization problem (21), while the latter one does not contain the oracle. Therefore, in the MPC-based controller, constraint (21)-(b) is replaced by

$$\tilde{\boldsymbol{x}}_k = \bar{\boldsymbol{x}}_k.$$

Moreover, a quantitative comparison is conducted using the following tracking performance index:

$$PI = \int_{t_0}^{t_1} e_y^2(t)dt.$$

The comparison results are shown in Fig. 9, while the performance indices are $PI_{MPC} = 380.04$ and $PI_{LBMPC} = 208.15$. It can be concluded that the LBMPC-base controller outperforms the MPC-based one regarding the tracking performance due to the oracle. However, the transient behavior of the LBMPC-base controller may be worse than that of the MPC-based one in some cases, for example, at 0.8×10^4 m and 2.1×10^4 m. Besides, the fluctuation of yaw rate of the LBMPC-based controller seems to increase. One possible reason is the design of the controller parameters. In fact, the weight matrix $\boldsymbol{Q}$ would affect the vehicle's behavior to some extent, which, in turn, would affect the driver's tenseness and interference in the vehicle. This type of disharmonious cooperation can be improved by further fine-tuning the parameters of the system model and the controllers.

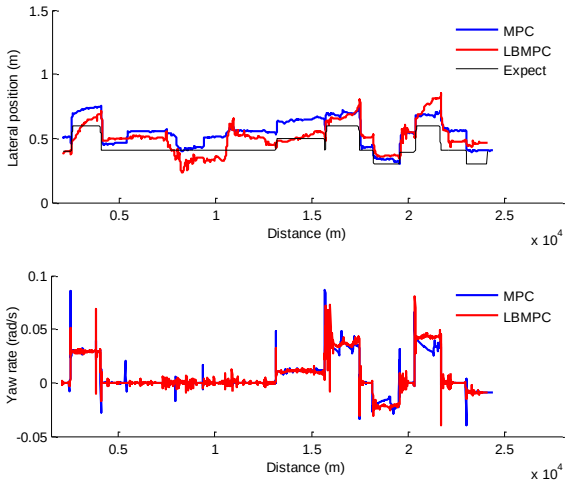


Fig. 9. Controller performance comparison

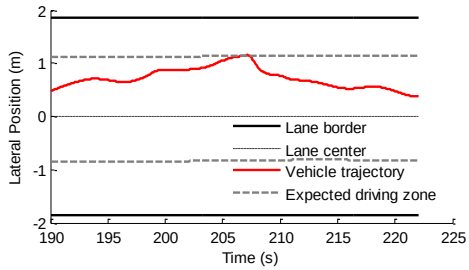
B. *Lane-Keeping Scenario*

In the lane-keeping scenario, the driver was required to stay within one of the lanes over a complete loop on the ring road. The LDP mode and the LK Co-Pilot mode were verified separately.

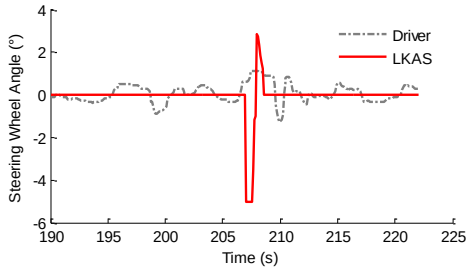
1) *LDP Mode*

The LDP function was activated 18 times during the driving test when unintended lane departures were detected. Moreover, four of them were overcome by the driver.

Fig. 10 shows the working process of the LDP mode during an observed lane departure case. The LKAS started intervening in the lane-keeping process when the vehicle's lateral position exceeded the range of the EDZ. In the first stage, a steering angle toward the lane center side was applied to reduce the lateral offset error, and then, the assistance steering angle was applied toward the opposite side to reduce the heading error in the second stage. The steering wheel angle of the LKAS remained zero before and after the interference since the LDP mode was not responsible for the lane-keeping task.



(a) Vehicle trajectory



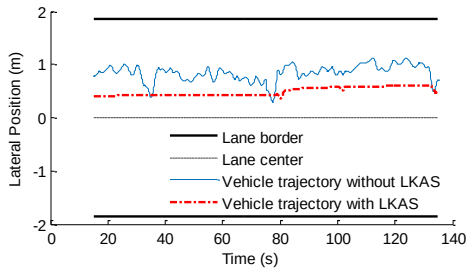
(b) Steering wheel angle

Fig. 10. The lane-keeping scenario (LDP mode)

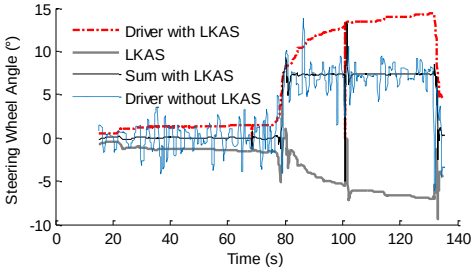
2) *LK Co-Pilot Mode*

The LK Co-Pilot mode was active throughout this test. A continuous data segment was selected to demonstrate its effectiveness. In the first 60 s of the segment, the road was straight, while in the next 60 s, the road was curved with a radius of 1 km.

As Fig. 11(a) shows, with the LK Co-Pilot system, the distribution of the vehicle's lateral offset was concentrated over an area closer to the lane center and the fluctuations were less erratic. An observable offset of about 0.5 m exists between the vehicle trajectory and the lane center. This is because the LKAS learned the driver's lane-keeping characteristics online and dynamically adjusted the control objective to drive in the left area of the lane.



(a) Vehicle trajectory



(b) Steering wheel angle

Fig. 11. The lane-keeping scenario (LK Co-Pilot mode)

As Fig. 11(b) shows, the driver corrected the steering wheel frequently without the LK Co-Pilot system. When the LK Co-Pilot system was switched on, the fluctuation frequency and range of the driver's steering angle decreased. This is because the LK Co-Pilot system compensated for the trajectory error by applying an opposite steering input to the vehicle according to the EDZ. Therefore, the driver did not need to correct the steering wheel frequently, and in this sense, the steering burden was reduced.

C. Lane-Change Scenario

To validate the effectiveness of the proposed LKAS in lane-change scenarios, other vehicles were designed to move in a way that may trigger the driver's lane-change intention. Moreover, the driver was required to decide whether and how to change lanes by himself. The LDP mode and the LK Co-Pilot mode were verified separately.

1) LDP Mode

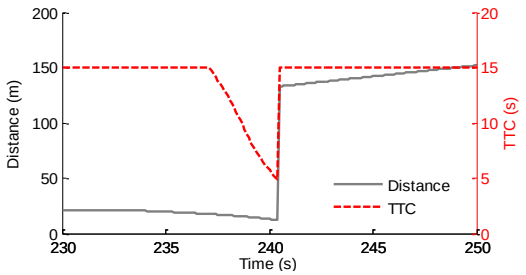
As Fig. 12(a) shows, since the preceding vehicle was moving slowly, the ego vehicle kept approaching the preceding vehicle and the time to collision (TTC) kept decreasing. To avoid a collision, the driver changed lanes at about 241 s. As shown in Fig. 12(b) and Fig. 12(c), when the ego vehicle's lateral position was close to the lane marking, the LDP system applied a steering input in the direction opposite to the driver's steering angle. In less than $T_o = 1$ s, the driver's real intention was recognized, and the LDP assistance was canceled. In this mode, the driver was still responsible for the lane-keeping task after the lane change. Therefore, the driver's steering angle fluctuated as well. This phenomenon was consistent with the decision-making strategy of the LDP mode shown in Fig. 3.

2) LK Co-Pilot Mode

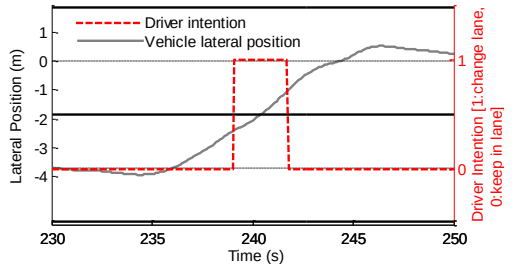
In Fig. 13(a), a decrease in the inter-vehicle distance and TTC was observed, which is similar to the one shown in Fig. 12(a). In this case, the driver changed lanes at about 169 s. As shown in Fig. 13(b) and Fig. 13(c), when the ego vehicle maintained a relatively stable lateral position, the LK Co-Pilot system was in the loop with the steering task at first. After the driver's lane-change intention was recognized, the LK Co-Pilot assistance was canceled until the lane change ended. Then, the LK Co-Pilot system provided the driver with steering control assistance again. When the lateral offset decreased to a very small value, the LK Co-Pilot system helped the driver to reduce the heading error by steering toward the right at about 170 s and the left at about 172 s. Thereafter, the vehicle remained in the EDZ, and the LK Co-Pilot system continued to provide lane-keeping assistance by steering slightly toward the right to compensate for the driver's steering angle. This demonstrates the effectiveness of LK Co-Pilot mode.

VI. Conclusion

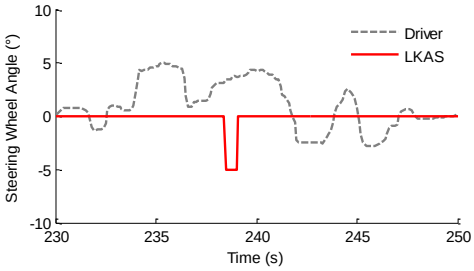
In this study, we proposed an advanced LKAS with two switchable assistance modes: an LDP mode and an LK Co-Pilot mode. For each of the two modes, two-stage decision-making strategies were proposed using the concept of EDZ, and control algorithms were designed using the LBMPC method. Driver-in-the-loop experiments on a driving simulator for both



(a) Distance and TTC



(b) Vehicle trajectory



(c) Steering wheel angle

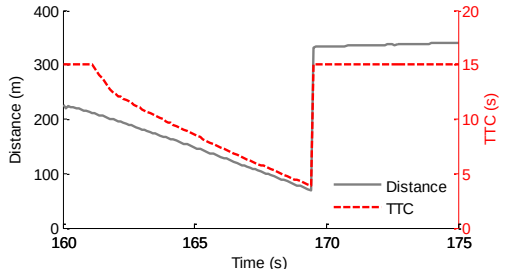
Fig. 12. The lane-change scenario test (LDP mode)

lane-keeping and lane-change scenarios were used to validate the effectiveness of both modes. Additionally, it was demonstrated that the LK Co-Pilot mode can further reduce driving burden in the sense of avoiding frequent steering correction.

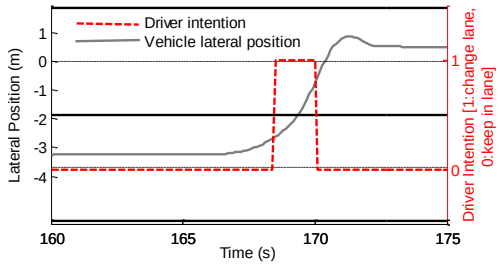
The proposed approach has the advantages of switchable modes and robustness. However, its application in real vehicles poses many challenges, *e.g.*, algorithm implementation in embedded processors and computational resource requirements, which need a tradeoff between performance and cost. Moreover, the parameters studied herein are determined partially through experience and intuition, which may not be the best. A rigorous parameter and threshold selection method deserves a systematic study when developing a product LKAS. Our future work will focus on experimental validation by involving human drivers with various driving characteristics. Furthermore, experiments on real vehicles need to be carried out to validate the applicability and performance of the proposed system.

Acknowledgment

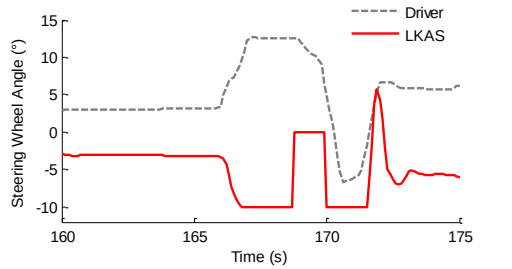
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(a) Distance and TTC



(b) Vehicle trajectory



(c) Steering wheel angle

Fig. 13. The lane-change scenario (LK Co-Pilot mode)

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